WEEK 2 · RESP-100 BASIC EKG INTERPRETATION

12-Lead EKGs

Walraven, *Basic Arrhythmias* 8th ed. · Appendix C: 12-Lead Electrocardiography · lecture 0:00–0:26

Why we do a 12-lead, how electrodes make leads, where the 10 electrodes go, what each of the 12 views looks like, and what a 12-lead can't tell you.

1 · Why a 12-lead? | 2 · What is a lead? | 3 · Rule of electrical flow | 4 · Bipolar vs unipolar | 5 · Einthoven's triangle (I, II, III) | 6 · Augmented leads | 7 · Electrode placement | 8 · Frontal & horizontal planes | 9 · The 12-lead printout | 10 · MI on a 12-lead | 11 · Limitations | Cheat sheet | Quiz

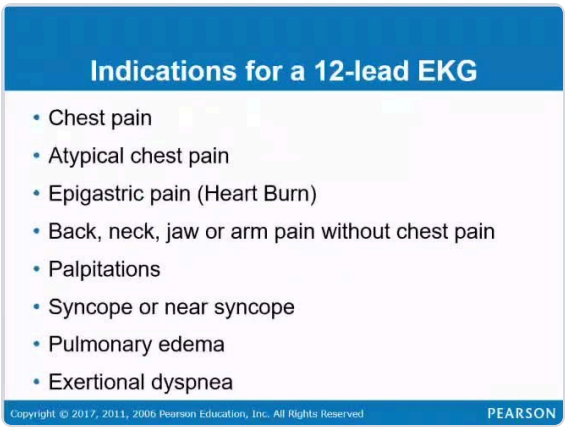
1 Why do a 12-lead EKG?

Big idea: a 12-lead records the heart's electrical activity from 12 angles. Its **#1 job is to diagnose a heart attack** = **myocardial infarction (MI)**.

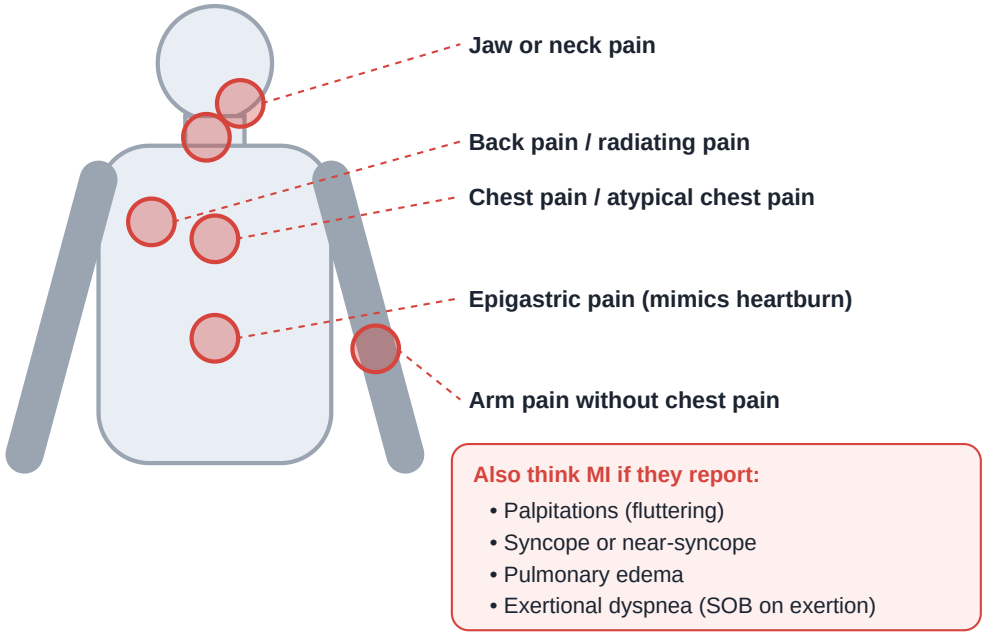
- Also called an **EKG** or **ECG**.
- Electrode patches on the skin of the **chest, arms and legs**.
- Assesses the electrical and muscular **functions** of the heart and checks for signs of heart disease.

LINK IT

MI = heart attack. If you see "MI" on the exam, think "heart attack → get a 12-lead".



Slide: indications for a 12-lead EKG.



What an MI patient may complain of. Pain can be anywhere on this map, even with NO chest pain.

WHY PULMONARY EDEMA? (0:01:39)

If the MI damages the **left ventricle**, it can't push blood forward. Fluid backs up into the lungs (pulmonary edema), so the patient is short of breath, especially on exertion.

2 What is a lead?

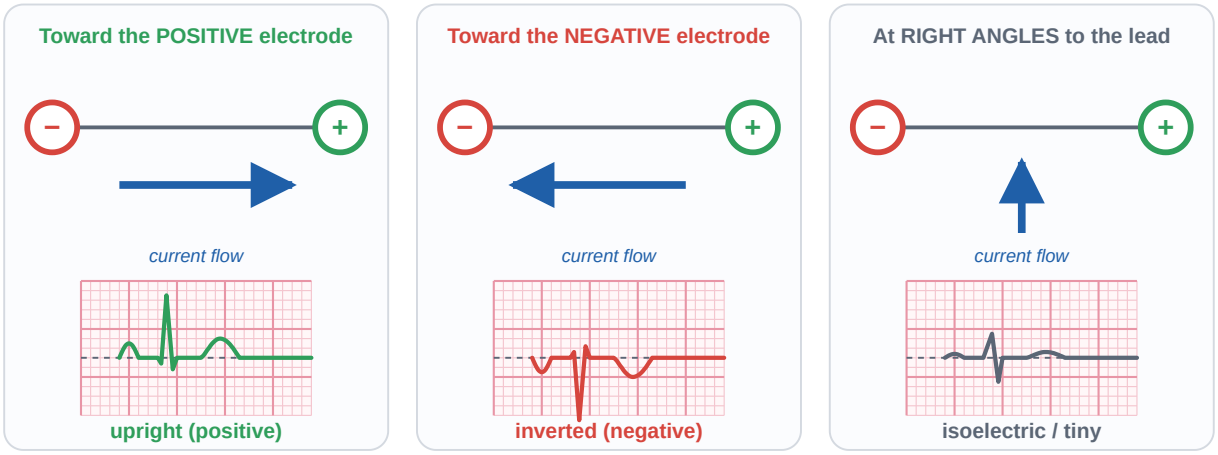
Big idea: a **lead** is a combination of electrodes. It records the flow of electricity **between two points** on opposite sides of the heart. Think of a lead as a **camera angle**.

ELECTRODE ≠ LEAD

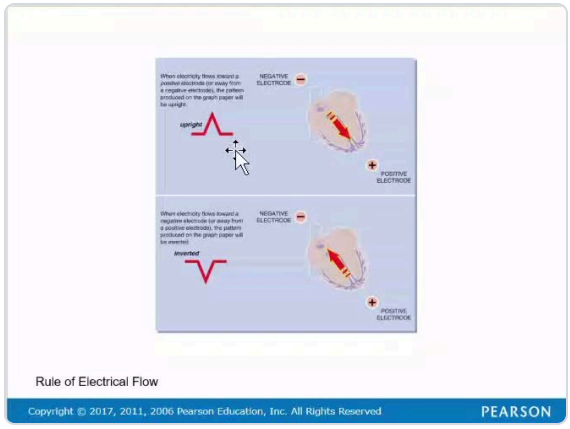
An **electrode** is the sticky patch. A **lead** is the view made by comparing electrodes. **10 electrodes → 12 leads (views).**

3 The rule of electrical flow

Big idea: toward **+** = **upright**; toward **−** = **downward**; at right angles = **isoelectric** (neither up nor down).



The same heartbeat looks different depending on which way the current travels relative to the lead.



Slide: current toward the positive electrode = upright; toward the negative electrode = inverted.

WHAT THE INSTRUCTOR STRESSED (0:03:29, 0:08:56)

The **isoelectric line** is the neutral line through the middle: above = positive, below = negative. The **right arm is the negative electrode**; the left arm and left leg are positive.

MEMORY TRICK

"**Right is wrong**": current heading to the **R**ight arm writes the wave upside-down.

4 Bipolar vs unipolar leads

Big idea: a **bipolar** lead has one + and one − electrode. A **unipolar** lead has one + electrode, and its "other end" is a **central terminal**: an electrically neutral point made by combining the other electrodes, in the centre of the heart.

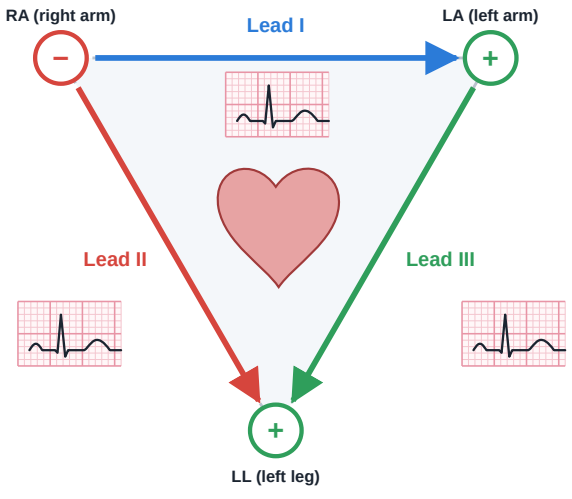
	Bipolar	Unipolar
Ends	one + and one − electrode	+ electrode vs. the central terminal
Leads	I, II, III (standard limb leads)	aVR, aVL, aVF (augmented) and V1–V6 (chest)
Picture	sides of Einthoven's triangle	spokes from the centre

HEADS-UP ON WORDING (0:05:34)

In class, "most of the leads on the chest" were called bipolar. By the slide's definition, only **I, II and III** are bipolar. The augmented leads (and, in the textbook, V1–V6) are **unipolar**.

5 Einthoven's triangle: leads I, II, III

Big idea: an imaginary **inverted equilateral triangle** formed by the **two shoulders and the pubis**, with the heart at the centre. Each side is one lead.



Every lead points from − to + ; arrows show the direction each lead "looks"

Lead I: RA → LA · Lead II: RA → LL · Lead III: LA → LL. All three record upright waves.

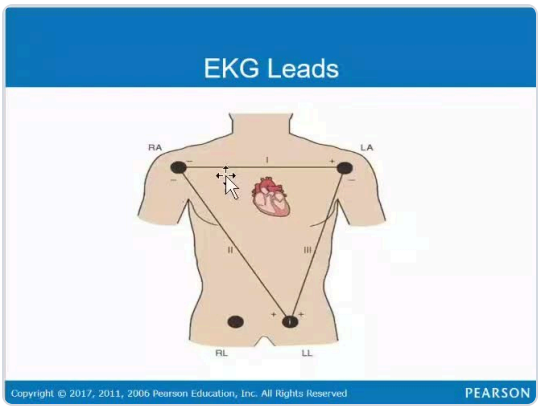
Lead	From (−) → To (+)
I	right arm → left arm across the top of the chest
II	right arm → left leg through the centre; the **standard monitoring lead**
III	left arm → left leg the lateral side

COUNT THE L'S

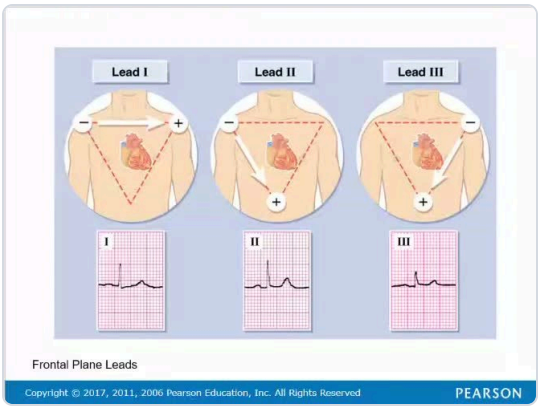
Count the L's in the electrode names: **I** = RA→**L**A (1 L) · **II** = RA→**LL** (2 L's) · **III** = **L**A→**LL** (3 L's).

FINE PRINT

In lead III the left arm is the **negative** end and the left leg is positive. Current still flows toward the + left leg, so lead III is upright.



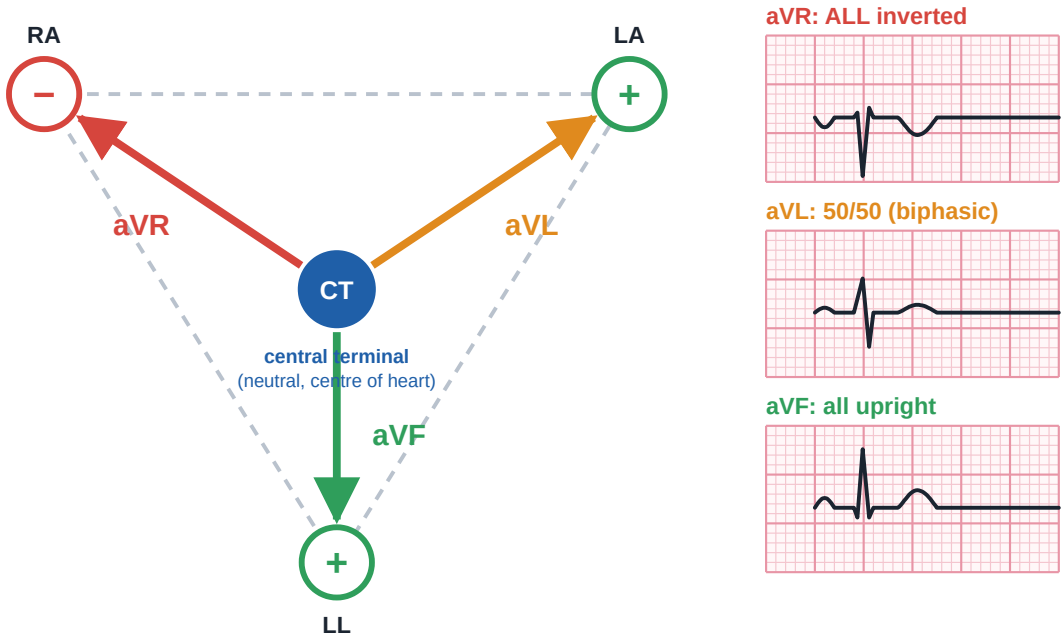
Slide: the triangle on a real person (RA, LA, LL; RL is the ground).



Slide: frontal-plane leads I, II, III: all upright P, QRS and T.

6 Augmented limb leads: aVR, aVL, aVF

Big idea: from the **central terminal (CT)** to one limb: **aVR** → **R**ight arm, **aVL** → **L**eft arm, **aVF** → **F**oot (left leg). These are the **unipolar** augmented leads.



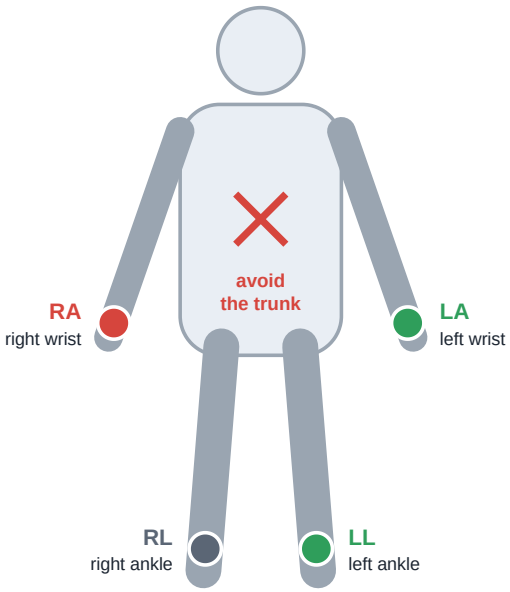
Left: CT to each limb. Right: what each lead records.

LIKELY EXAM POINT (0:18:17–0:19:31)

aVR is all negative: the inverted P, QRS and T all head toward the negative right arm. **aVL is about 50/50** (part up, part down). **aVF is all positive.**

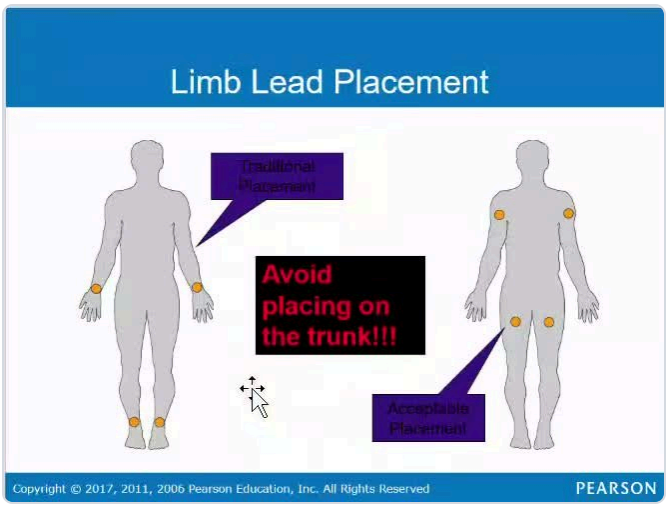
7 Electrode placement: 4 limbs + 6 chest = 10

Big idea: limb electrodes go on the **wrists and ankles** (not the trunk). Chest electrodes V1–V6 go in exact **intercostal spaces** (spaces between the ribs).

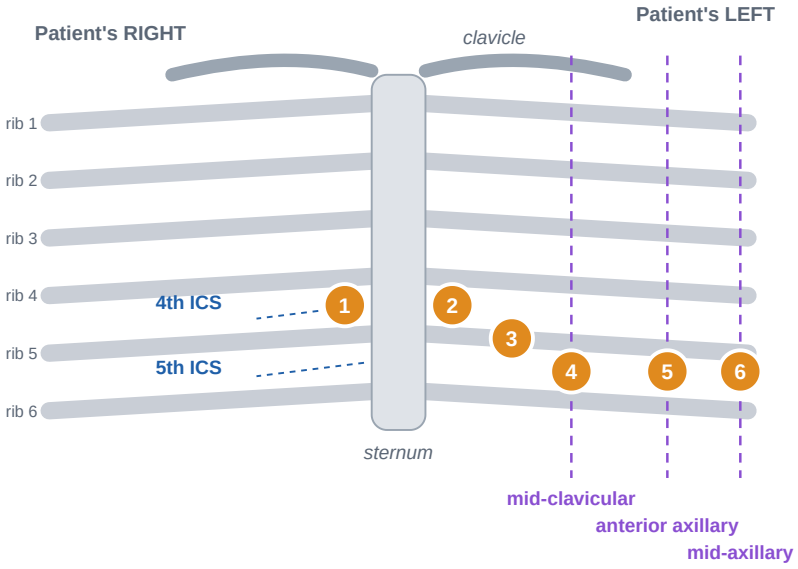


Patient's right is on YOUR left.

Limb leads: RA, LA (wrists), RL, LL (ankles). Move them up the limb only if you must (amputation, cast).



Slide: traditional placement (wrists/ankles) vs. acceptable placement. Avoid the trunk!



Chest (precordial) leads. Orange dots = electrodes. Place V4 before V3.

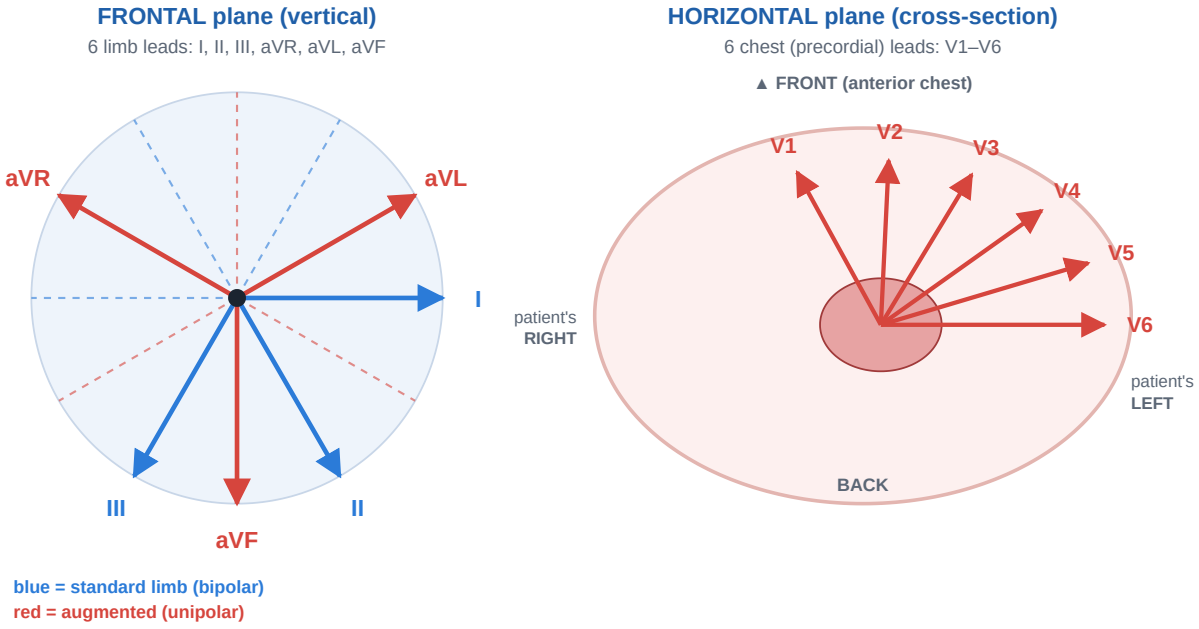
Lead	Spot	How to find it
V1	4th ICS, **right** of sternum	count down 4 rib spaces; the patient's right, not yours
V2	4th ICS, **left** of sternum	mirror image of V1
V4	5th ICS, left **mid-clavicular** line	straight down from the middle of the collarbone. **Skip V3 and do V4 next**
V3	halfway between V2 and V4	easy once V2 and V4 are on
V5	5th ICS, **anterior axillary** line	front edge of the armpit, level with V4
V6	5th ICS, **mid-axillary** line	lift the arm; middle of the armpit, level with V4

MEMORY TRICK

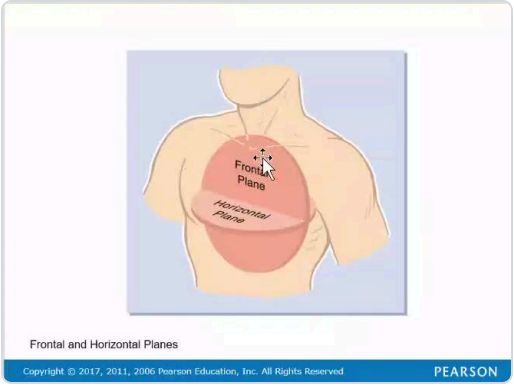
"**4-4-5-between-5-5**": V1 & V2 in the 4th space, V4–V6 in the 5th, V3 in between. Axillary = **armpit**; clavicular = **collarbone**.

8 Frontal & horizontal planes

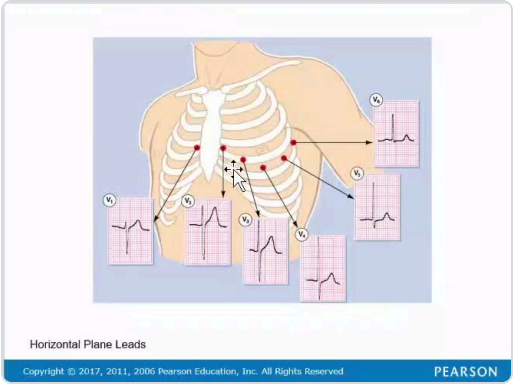
Big idea: the 6 **limb leads** (I, II, III, aVR, aVL, aVF) look at the **frontal (vertical)** plane. The 6 **chest leads** (V1–V6) look at the **horizontal** plane.



Two planes, 12 views. Arrows point toward each lead's positive pole.



Slide: the frontal plane (up and down) and the horizontal plane (around the chest).



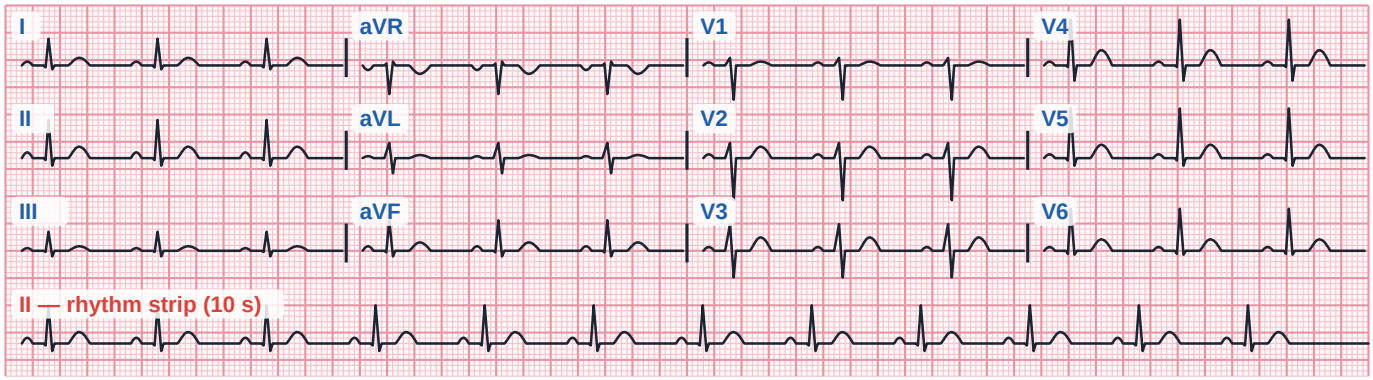
Slide: a snippet from each chest lead, V1 to V6.

TRANSCRIPT SLIP (0:19:36)

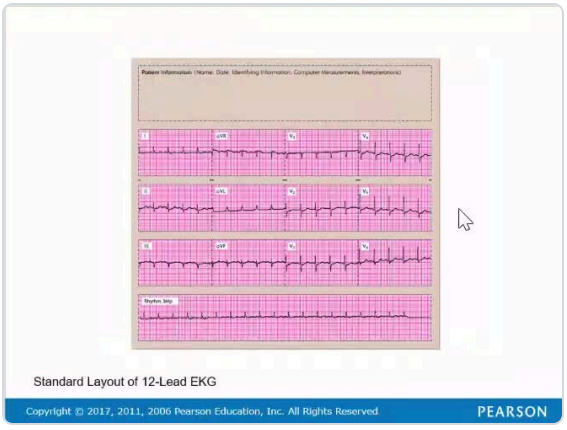
At one point the lecture says the "augmented leads and pre-cordial leads" look at the horizontal plane. The slides, and the Week 3 review (0:13:35), are clear: **augmented leads = frontal plane**, **precordial = horizontal plane**.

9 The standard 12-lead printout

Big idea: four columns of short snippets (I-II-III, aVR-aVL-aVF, V1-V2-V3, V4-V5-V6) and a long **Lead II rhythm strip** across the bottom.



Drawn example of a normal 12-lead. Note aVR (all inverted), aVL (biphasic), and the R wave growing from V1 (small r, deep S) to V6 (tall R).



Slide: standard layout of a 12-lead EKG.

REMEMBER (0:20:10)

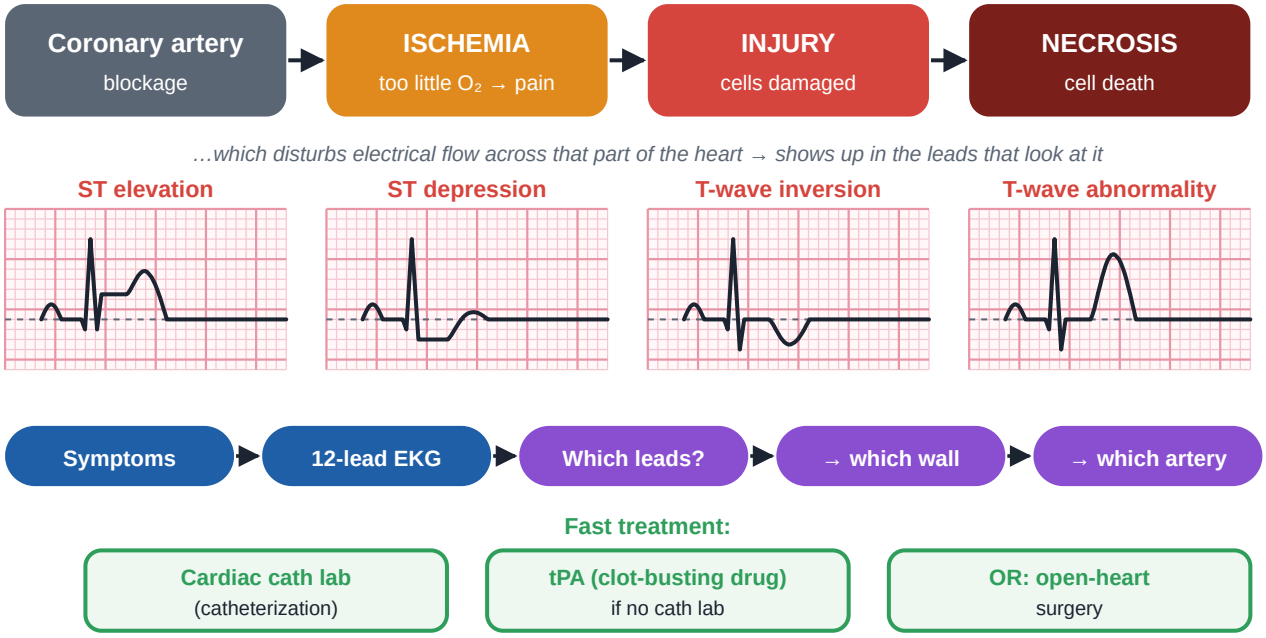
The rhythm strip at the bottom is **always recorded in Lead II**.

READ IT LIKE A BOOK

Start at the **top-left** (lead I), go down each column, then move right. Each column is a snapshot of about 2.5 s.

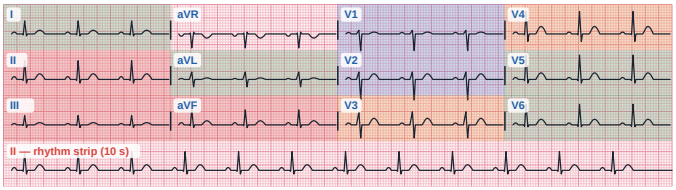
10 Finding an MI on the 12-lead

Big idea: a blocked coronary artery starves one area of muscle. That area's electrical flow changes, and the change shows up in the **leads that look at that area**.



From blockage to treatment. On the 12-lead, look for ST elevation, ST depression, and T-wave abnormalities or inversion.

Which leads look at which wall?



Colour = wall of the heart. Red: inferior · green: lateral · blue: septal · orange: anterior.

Wall	Leads
Inferior	II, III, aVF
Lateral	I, aVL, V5, V6
Septal	V1, V2
Anterior	V3, V4

BEYOND THIS LECTURE

The lecture gave the anterior wall as an example: look at the leads covering it, then work out which coronary artery feeds it. The full map above is the standard textbook map (Appendix C). Learn it now, and you'll use it later.

11 Limitations of a 12-lead

Big idea: an EKG shows **electricity, not pumping**, and only **at the moment it was recorded**.

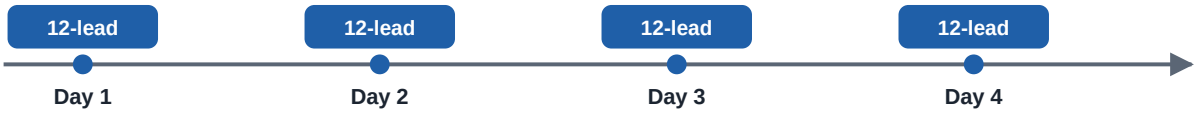
✓ EKG CAN show
ELECTRICAL status



✗ EKG CANNOT show
the MECHANICAL pump

To check the pump:
- feel for a pulse
- take a blood pressure
- central line: RA, PA & capillary pressures; cardiac output

A 12-lead is a SNAPSHOT: it shows the moment it was recorded, not the future



After an MI: SERIAL 12-leads (often daily) track progress or regression

Two limits: electrical only, and a snapshot in time. That's why MI patients get serial 12-leads.

WHAT THE INSTRUCTOR STRESSED (0:24:15–0:25:35)

To assess the **mechanical pump**: feel a pulse, take a BP, or use an invasive central line (right-atrial, pulmonary-artery and pulmonary-capillary pressures; cardiac output). After treatment for an MI, patients get a **12-lead every day** to track how the MI is progressing or regressing.

★ Cheat sheet: Week 2 on one page

The 12 leads

Lead	From → to	Type	Looks like
I	RA → LA	bipolar	upright
II	RA → LL	bipolar	upright (monitor & rhythm strip)
III	LA → LL	bipolar	upright
aVR	CT → right arm	unipolar	**all inverted**
aVL	CT → left arm	unipolar	50/50 biphasic
aVF	CT → left leg (foot)	unipolar	upright
V1–V6	chest	unipolar*	R grows V1 → V6

*textbook classification; see the bipolar/unipolar note in section 4.

Rule of electrical flow

Toward +	upright
Toward − (right arm)	downward / inverted
Right angle to lead	isoelectric

Planes

Frontal (vertical)	I, II, III, aVR, aVL, aVF
Horizontal	V1–V6

Electrode placement (10)

Limbs	R & L wrists, R & L ankles (not the trunk)
V1	4th ICS, right sternal border
V2	4th ICS, left sternal border
V3	between V2 & V4 (place after V4)
V4	5th ICS, left mid-clavicular
V5	5th ICS, anterior axillary
V6	5th ICS, mid-axillary

MI essentials

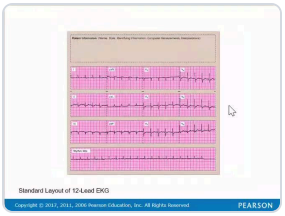
Main reason for a 12-lead	diagnose MI (heart attack)
Symptoms	chest/atypical/epigastric pain; back, neck, jaw, arm pain; palpitations; syncope; pulmonary edema; exertional dyspnea
Sequence	blockage → ischemia → injury → necrosis
Look for	ST elevation, ST depression, T-wave changes/inversion
Treatment	cath lab · tPA · open-heart surgery

Limitations

Electrical only	pump: pulse, BP, central line
Snapshot only	serial (daily) 12-leads

? Self-test quiz

1. What is the single biggest reason to perform a 12-lead EKG? What is the proper name for a heart attack?
2. List four symptoms that might make you get a 12-lead, including one that involves *no* chest pain.
3. Define a **lead**. How many electrodes create how many leads?
4. Current flows toward the right-arm electrode. Is the wave upright or inverted? What if the current runs at a right angle to the lead?
5. Fill in: Lead I = ___ → ___; Lead II = ___ → ___; Lead III = ___ → ___.
6. What is the "central terminal", and which leads use it?
7. **Identify the lead:** every wave (P, QRS and T) is below the isoelectric line. Which augmented lead is this most likely to be, and why?
8. Where exactly do V1, V2, V4, V5 and V6 go? Why is V3 placed last?
9. Your patient has a cast on the left leg. Where can you place the LL electrode?
10. Which leads look at the frontal plane, and which look at the horizontal plane?
11. **Read the printout:** on a standard 12-lead, what is in the top-left box, the bottom-right box, and the long strip at the bottom?



Standard 12-lead layout.

12. Put in order: necrosis, blockage, ischemia, injury. Name two 12-lead findings you'd look for.
13. Name the three fast treatments for an MI mentioned in the lecture.
14. Why can a normal-looking EKG NOT prove the heart is pumping well? Why do MI patients get daily 12-leads?

Answers. 1) To diagnose a myocardial infarction (MI). 2) Any four of: chest pain, atypical chest pain, epigastric pain ("heartburn"), back/neck/jaw/arm pain without chest pain (the no-chest-pain one), palpitations, syncope or near-syncope, pulmonary edema, exertional dyspnea. 3) A combination of electrodes that records electrical flow between two points on opposite sides of the heart; 10 electrodes → 12 leads. 4) Inverted (the right arm is negative); isoelectric. 5) I = RA → LA; II = RA → LL; III = LA → LL. 6) An electrically neutral point in the centre of the heart, made by combining electrodes; used by the unipolar augmented leads aVR, aVL, aVF. 7) aVR: the current flows toward the negative right arm, so everything is inverted. 8) V1 4th ICS right sternal border; V2 4th ICS left sternal border; V4 5th ICS left mid-clavicular; V5 5th ICS anterior axillary; V6 5th ICS mid-axillary. V3 goes halfway between V2 and V4, so V4 must be on first. 9) Move it higher up the leg (toward the thigh/base of the leg), keeping it on the limb and off the trunk. 10) Frontal: I, II, III, aVR, aVL, aVF. Horizontal: V1–V6. 11) Top-left = lead I; bottom-right (of the 4 columns) = V6; bottom strip = lead II rhythm strip. 12) Blockage → ischemia → injury → necrosis; ST elevation, ST depression, T-wave abnormality/inversion. 13) Cardiac catheterization (cath lab), tPA, open-heart surgery (OR). 14) The EKG shows electrical activity only, not the mechanical pump (check pulse/BP); it's a snapshot, so serial 12-leads track the MI's progress.